

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Buvanesh Y
Register Number	192372367

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

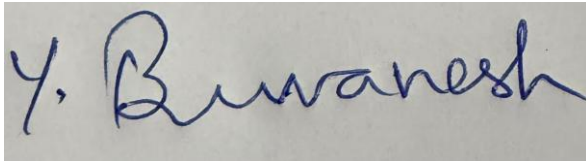
Duration : 60 Minutes

Code of Conduct

I **Buvanesh Y (192372367)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org , over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

EXP NO 1: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR LIBRARY BOOK RESERVATION SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SaaS

AIM:

To create a simple cloud software application for Library Book Reservation System using Zoho Creator as a cloud service provider to demonstrate Software as a Service (SaaS).

PROCEDURE:

Step 1: Go to zoho.com.

Step 2: Log in to the Zoho account.

Step 3: Select Zoho Creator from the available applications.

Step 4: Create a new application and enter the application name as SIMATS Library Book Reservation System.

Step 5: Create the new application for managing library books.

Step 6: Select Form to create a book details form.

Step 7: Add the necessary fields to the form such as:

- Title of the Book
- Author
- Publication
- Year
- Edition
- No. of Copies
- Rack No.
- Status (Taken/Returned)

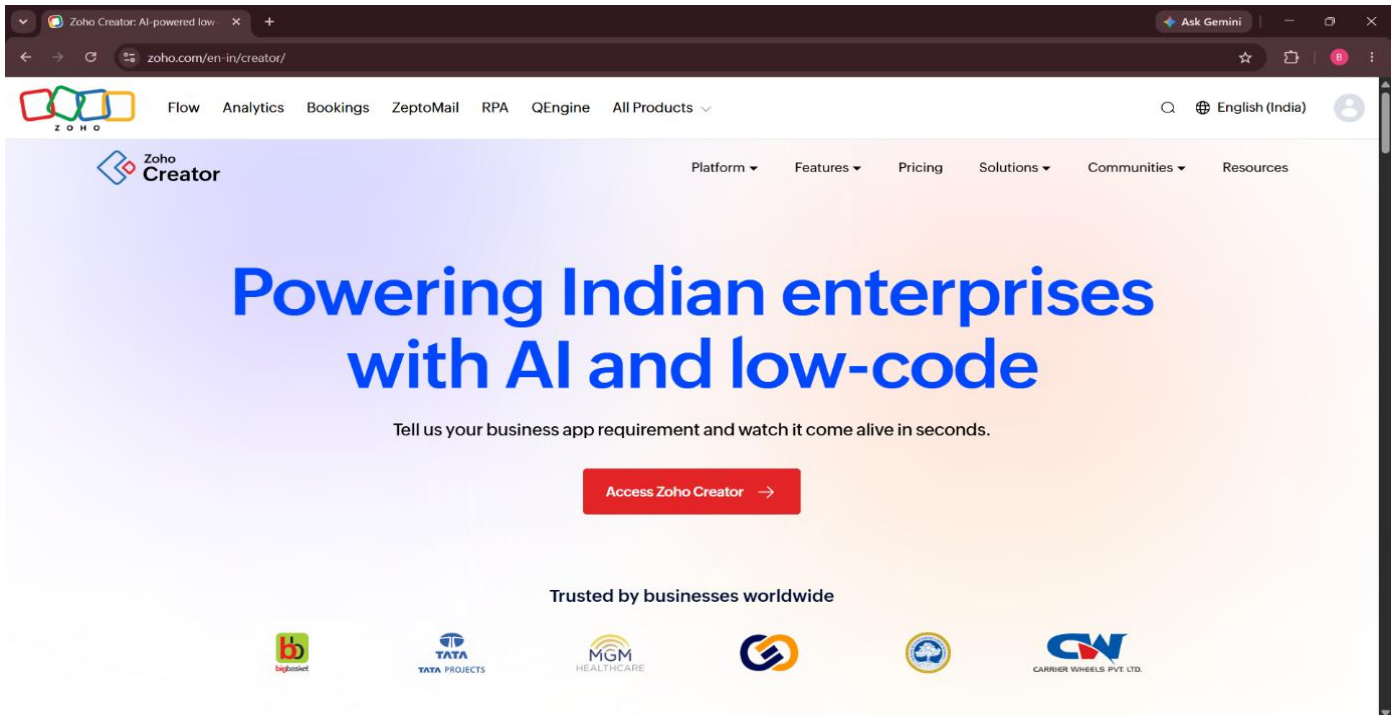
Step 8: Save the form and enter sample book details.

Step 9: Create a report to display all the available book details and their current status.

Step 10: The cloud-based library reservation software is created successfully and can be accessed through the internet.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

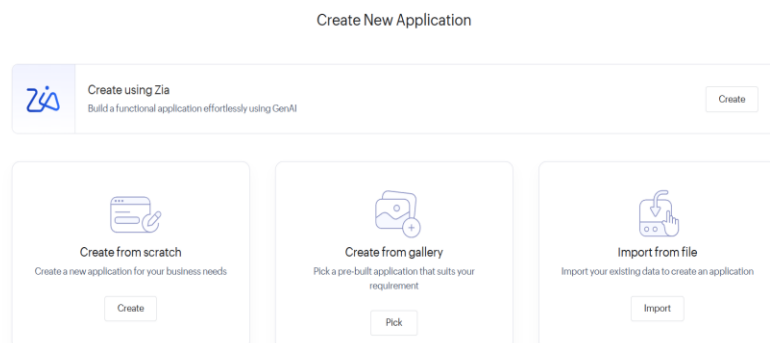


STEP 2: LOGINTOTHE ZOHO.COM



STEP 3: SELECT ONE APPLICATION

❖ Click Create from Scratch



STEP 4: ENTER APPLICATION NAME

✕


Create from scratch

Application Name *

☐ Enable Environment [Learn More](#)

Create Application

STEP 5: CREATED NEW APPLICATION

creator.zoho.in/appbuilder/buva1636/library-book-reservation-system/page/DashBoard/edit

✕

Create New Component

Create your component from scratch


Form


Report


Page


Workflow

Chats Contacts Here is your Smart Chat (Ctrl+Space) Trial expires in 6 days | [Upgrade](#) Get started faster! [Help](#)

❖ Create Multiple Forms according to our application.

STEP 6: THE SOFTWARE HASE BEEN CREATED.

Library Book Reservation System

Dashboard

Book Author Relations

Author Details

Book Catalog

6 days

Upgrade

Edit this application

Help

Buvenesh Y

Enter Book Title

Search

Total Total Copies

455.00

Total Book Author Relations

5

Total Author Details

5

Count of Book Catalog

5

Role

1 Authors

3 Authors

1 Editors

5 Total

All Book Author Relations

Book Title	Author Name	Role
Quantum Horizons	J.K. Rowling George R.R. Martin Stephen King J.R.R. Tolkien	Co Author
Quantum Horizons Green Earth Rising Midnight in Paris Code of the Future	J.K. Rowling George R.R. Martin Stephen King J.R.R. Tolkien Neil Gaiman	Author
The Silent Echo Quantum Horizons	J.K. Rowling George R.R. Martin	Co Author

Library Book Reservation System

Dashboard

Book Author Relations

Author Details

Book Catalog

6 days

Upgrade

Edit this application

Help

Buvenesh Y

Book Author Relation

Book Title

Author Name

Role

Submit

Reset

Author Name	Nationality	Birth Year	Biography	Contact Email	Contact Phone	Website URL	Awards
Brijesh	Pakistani	2006	Waste Fellow	iammad@gmail.com	+911234567890	https://creatorapp.zoho.in/buva1636/library-book-reservation-system#Report:book_author_relation_Report	None
Buvanesh	Indian	16032006	The Greate Author	buva1636@gmail.com	+918015979492	https://creatorapp.zoho.in/buva1636/library-book-reservation-system#Report:book_author_relation_Report	All National Awards

Showing 2 of 2

RESULT:

Thus, the **SIMATS Library Book Reservation System** was successfully created using **Zoho Creator** as a cloud service provider, demonstrating the **SaaS model**. The application allows library book details to be stored, managed, and accessed through the cloud.

EXP NO 2: CREATE A SIMPLE CLOUD SOFTWARE APPLICATION FOR PAYROLL PROCESSING SYSTEM USING ANY CLOUD SERVICE PROVIDER TO DEMONSTRATE SaaS

AIM:

To create a simple cloud software application for **Payroll Processing System** using **Zoho Creator** as a cloud service provider to demonstrate **Software as a Service (SaaS)**.

PROCEDURE:

Step 1: Go to **zoho.com**.

Step 2: Log in to the Zoho account.

Step 3: Select **Zoho Creator** from the available applications.

Step 4: Create a new application and enter the application name as **Payroll Processing System**.

Step 5: Create the new application for managing employee payroll details.

Step 6: Select **Form** to create an employee payroll details form.

Step 7: Add the necessary fields to the form such as:

- Name of the Employee
- Employee ID
- Experience in Present Organization
- Overall Experience
- Basic Pay
- HRA
- DA
- TA
- Gross Pay
- Net Pay

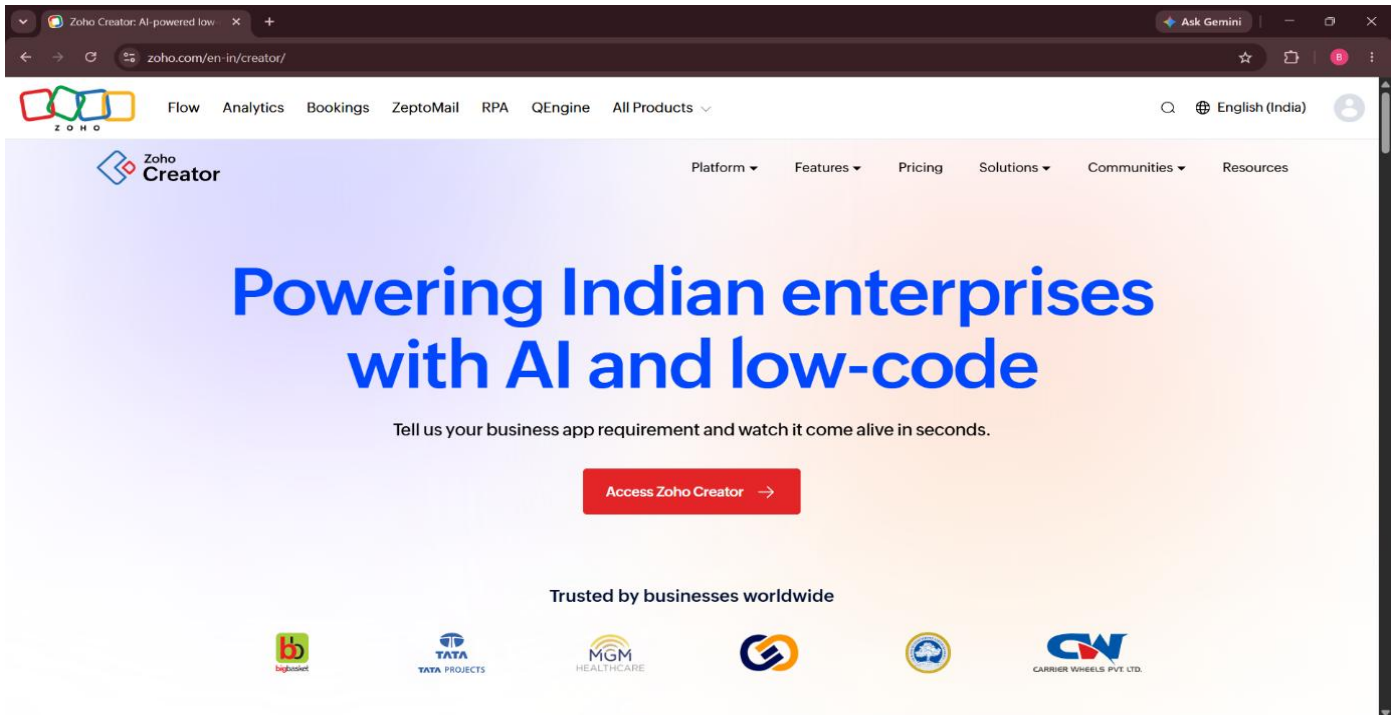
Step 8: Save the form and enter sample employee payroll details.

Step 9: Create a report to display the employee details and salary information.

Step 10: The cloud-based Payroll Processing System is created successfully and can be accessed through the internet.

IMPLEMENTATION:

STEP1: GOTO ZOHO.COM

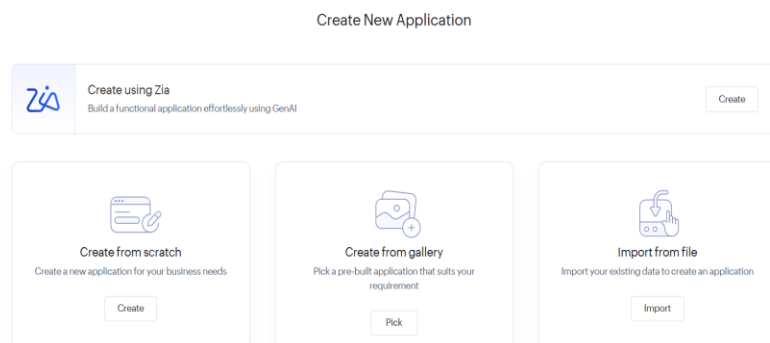


STEP 2: LOGINTOTHE ZOHO.COM

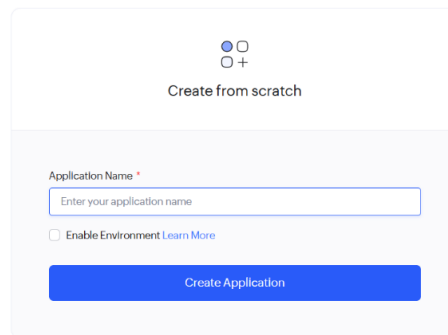


STEP 3: SELECT ONE APPLICATION

❖ Click Create from Scratch

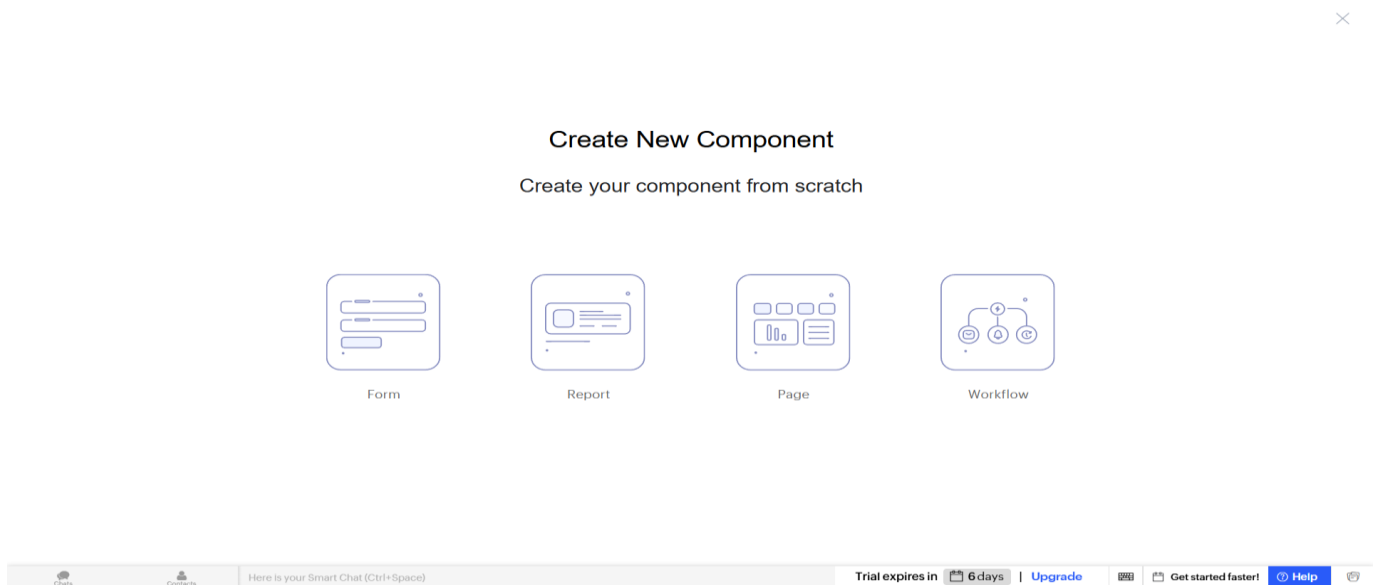
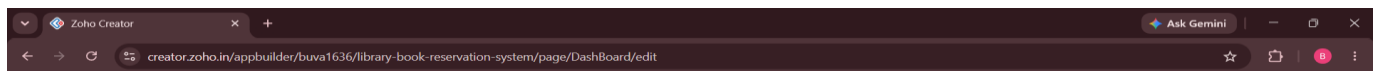


STEP 4: ENTER APPLICATION NAME



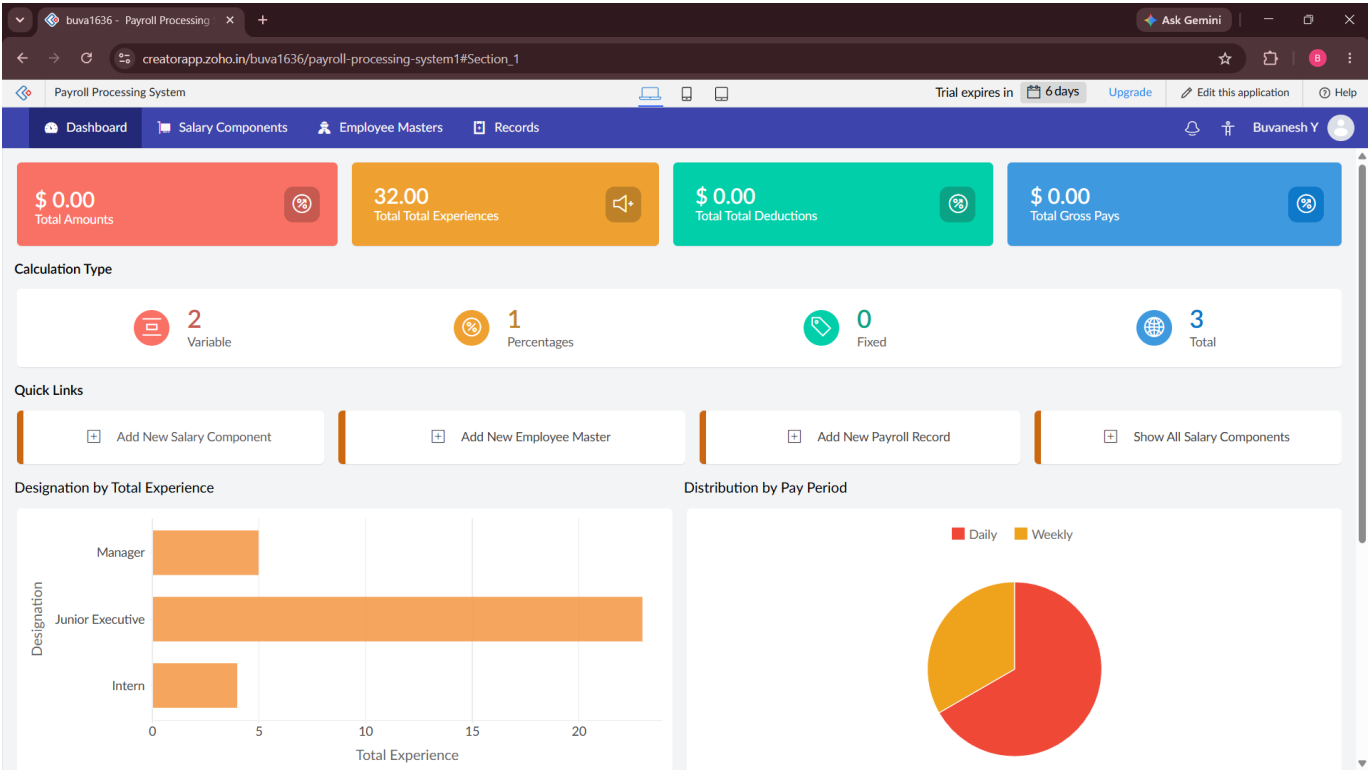
The screenshot shows a modal window titled "Create from scratch" with a close button (X) in the top right corner. Inside the modal, there is a section labeled "Application Name *" with a text input field containing the placeholder "Enter your application name". Below the input field is a checkbox labeled "Enable Environment" with a link "Learn More" next to it. At the bottom of the modal is a blue button labeled "Create Application".

STEP 5: CREATED NEW APPLICATION



❖ Create Multiple Forms according to our application.

STEP 6: THE SOFTWARE HASE BEEN CREATED.



The screenshot shows the 'Salary Component' form. It includes fields for Component Name, Component Code, Amount (with a currency symbol), Is Taxable (checkbox), Employee (dropdown), Pay Period (dropdown), Calculation Type (dropdown), and Remarks (text area). At the bottom, there are Submit and Reset buttons. A language selection dropdown is also visible, showing English (India) as the selected option.

Salary Component

Component Name:

Component Code:

Amount: \$

☐ Is Taxable

Employee:

Pay Period:

Calculation Type:

Remarks:

English (India)
English (India)
To switch input methods, press Windows key + space.

buva1636 - Payroll Processing

creatorapp.zoho.in/buva1636/payroll-processing-system1#Report:employee_master_Report

Payroll Processing System

Trial expires in 6 days Upgrade Edit this application Help

Dashboard Salary Components Employee Masters Records

All Employee Masters

Employee Name	Total Experience	Email Address	Phone Number	Address	Designation	Department	Employment Status	Joining Date
Buvanesh Y	10	buva1636@gmail.com	+91451304770	21 DUNPHY ST	Senior Executive	IT	Active	15-Apr-2026

Showing 1 of 1

buva1636 - Payroll Processing

creatorapp.zoho.in/buva1636/payroll-processing-system1#Report:salary_component_Report

Payroll Processing System

Trial expires in 6 days Upgrade Edit this application Help

Dashboard Salary Components Employee Masters Records

All Salary Components

Component Name	Component Code	Amount	Is Taxable	Employee	Pay Period	Calculation Type	Remarks
Saveetha University	16032006	\$ 10,000,000.00	true	Buvanesh Y	Daily	Percentage	Good

Showing 1 of 1

Show Summary

RESULT:

Thus, the **Payroll Processing System** was successfully created using **Zoho Creator** as a cloud service provider, demonstrating the **SaaS model**. The application allows employee salary details to be stored, processed, and accessed through the cloud.

EXP NO 3 :Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

DATE:

AIM:

To demonstrate virtualization by installing type-2 hypervisor in your device, create and configure VM image with a host operating system (either windows/linux).

PROCEDURE:

STEP 1:Download VMware workstation and installed as type 2hypervisor.

STEP2:Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

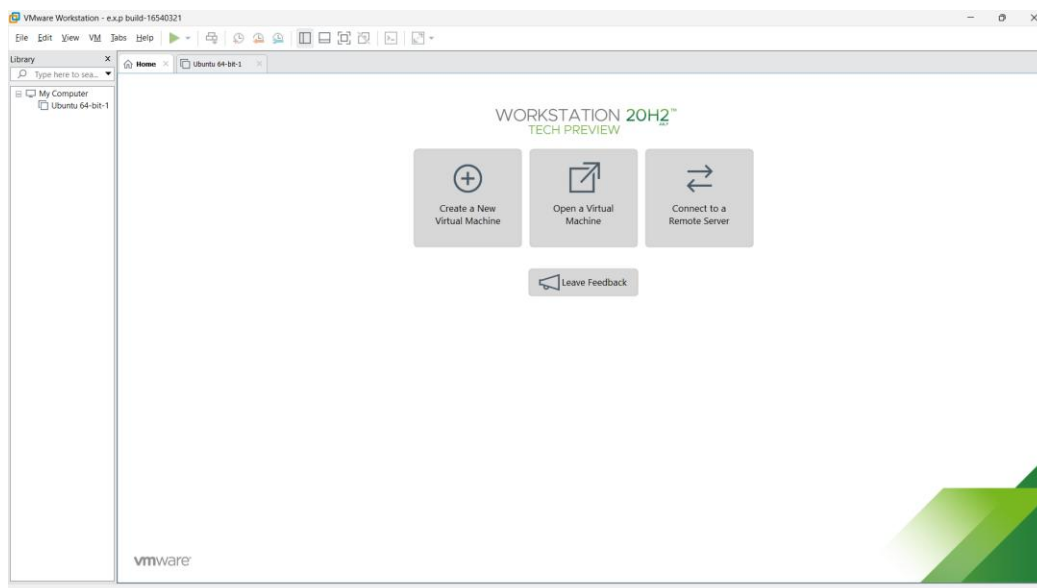
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

IMPLEMENTATION:

STEP 1:DOWNLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2HYPERVISOR

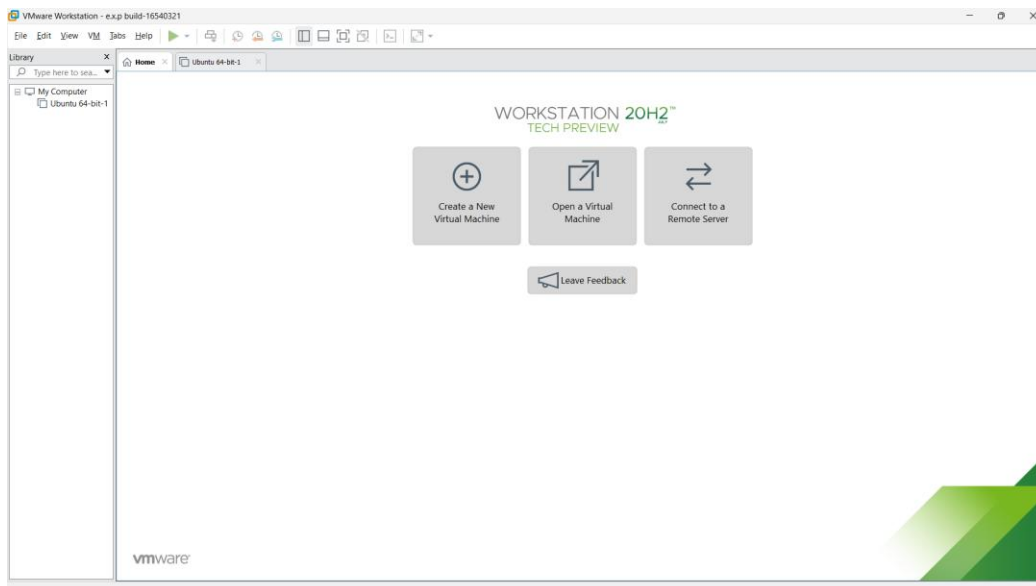


STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

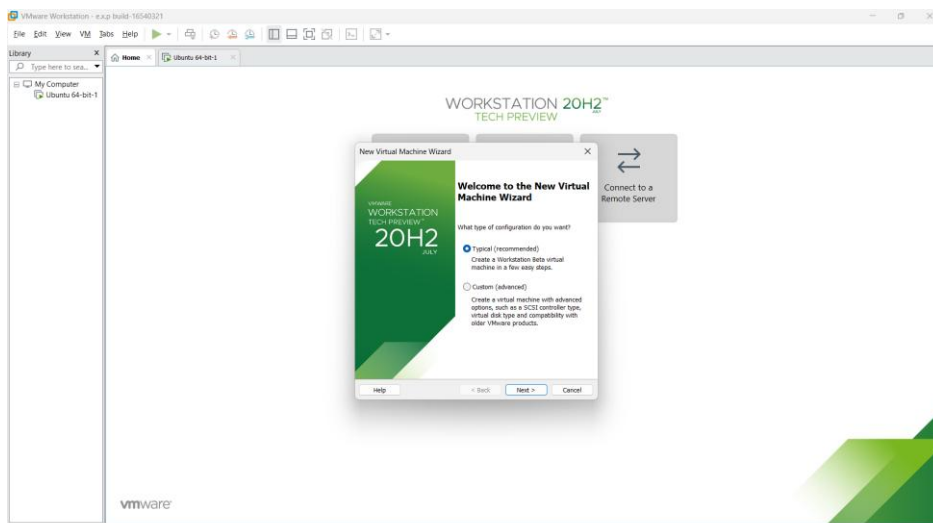
Index of /11.x/x86/release/

../	09-Feb-2020 11:50	-
distribution_files/	03-Dec-2019 11:14	-
src/	01-Apr-2020 07:49	14757888
Core-11.1.iso	01-Apr-2020 07:49	48
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	50639
Core-11.1.iso.zsync	01-Apr-2020 07:49	14757888
Core-current.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso	01-Apr-2020 07:50	52
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	369358
CorePlus-11.1.iso.zsync	01-Apr-2020 07:50	216006656
CorePlus-current.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso	01-Apr-2020 07:50	52
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	68301
TinyCore-11.1.iso.zsync	01-Apr-2020 07:50	19922944
TinyCore-current.iso		

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM



STEP 4: DO THE BASIC CONFIGURATION SETTINGS.



WORKSTATION 20H2™ TECH PREVIEW

New Virtual Machine Wizard

Guest Operating System Installation

A virtual machine is like a physical computer; it needs an operating system. How will you install the guest operating system?

Install from:

☐ Installer disc:

No drives available

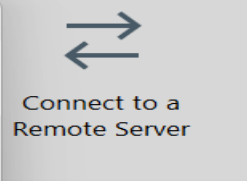
☒ Installer disc image file (iso):

D:\10.CLOUD SOFTWARE\ubuntu-18.04.3-desktop-ar Browse...

Ubuntu 64-bit 18.04.3 detected.
This operating system will use Easy Install. [\(What's this?\)](#)

☐ I will install the operating system later.
The virtual machine will be created with a blank hard disk.

Help < Back Next > Cancel



Ubuntu 64-bit-1 - VMware Workstation - e.x.p build-16540321

File Edit View VM Tabs Help

Library

Type here to search

My Computer

Ubuntu 64-bit-1

Home x Ubuntu 64-bit-1 x

Ubuntu 64-bit-1

Power on this virtual machine

Edit virtual machine settings

Devices

Memory	2.0 GB
Processors	1
Hard Disk (SCSI)	15 GB
CD/DVD (SATA)	Auto detect
Network Adapter	NAT
USB Controller	Present
Sound Card	Auto detect
Printer	Present
Display	Auto detect

Description

Type here to enter a description of this virtual machine.

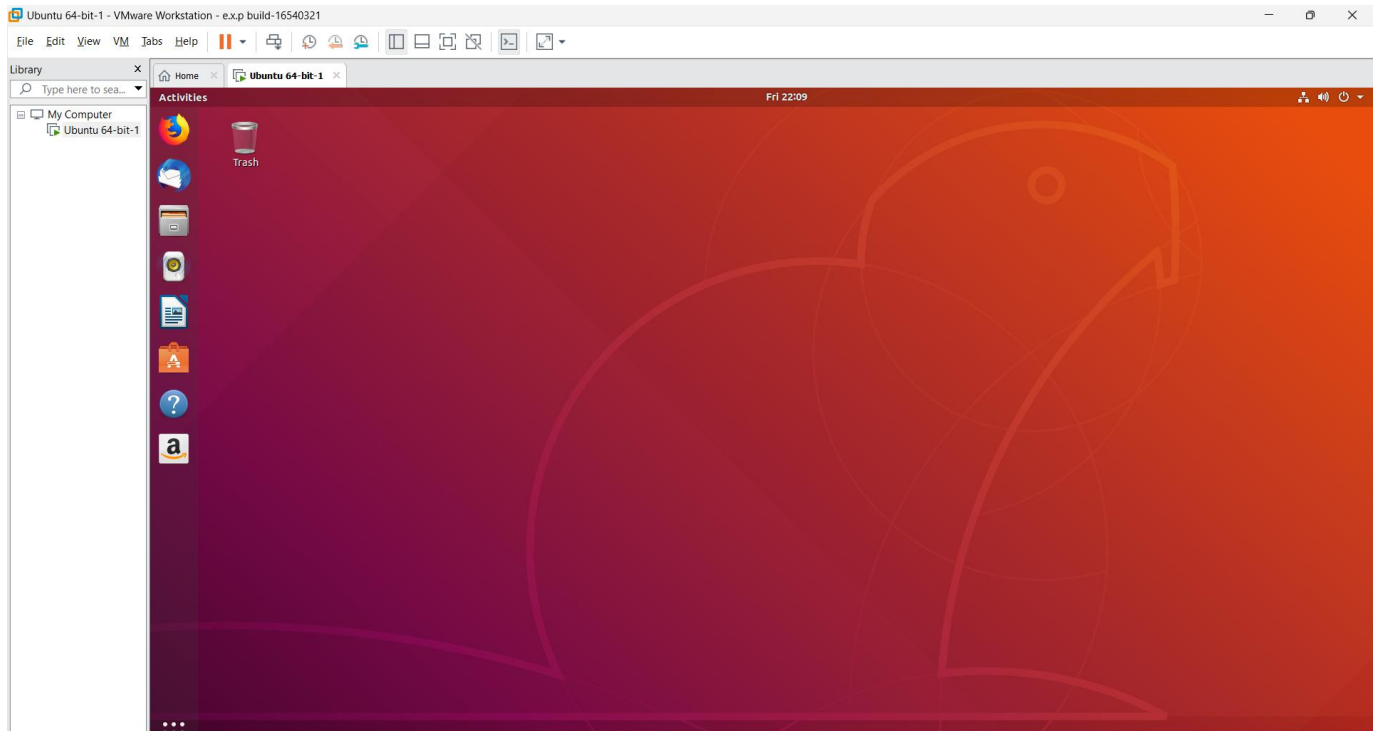
Virtual Machine Details

State: Powered off

Configuration file: C:\Users\BUVANESH\OneDrive\Documents\Virtual Machines\Ubuntu 64-bit-1\Ubuntu 64-bit-1.vmx

Hardware compatibility: Workstation Beta virtual machine

Primary IP address: Network information is not available



Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

AIM:

To create a snapshot of a vm and test it by loading the previous version/cloned vm

PROCEDURE:

STEP 1: GOTO VMWARE WORKSTATION.

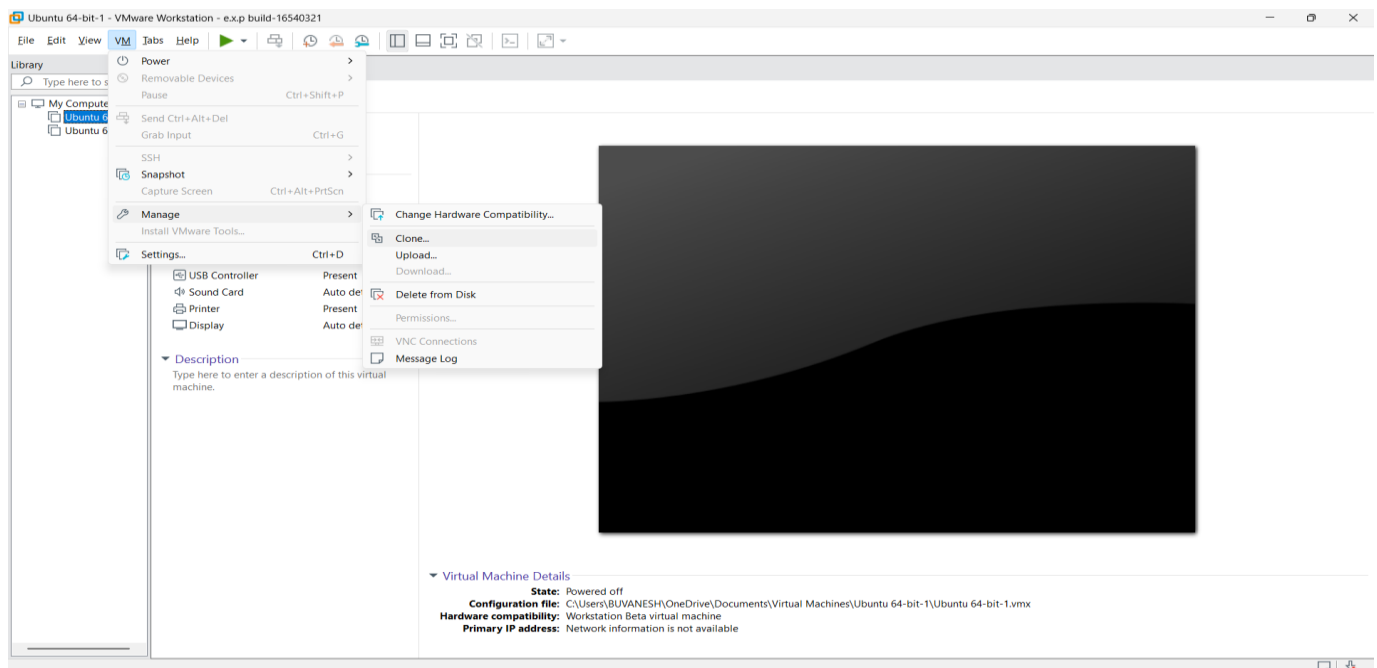
STEP 2: CREATE FILES ON DESKTOP.

STEP 3: CLICK ON VM AND SELECTS SNAPSHOT-> TAKE SNAPSHOT.

STEP 4: SNAPSHOT IS BEING DONE

IMPLEMENTATION:

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE



STEP 2: CLICK CLONE

